

GREAT FALLS, MT, USA

WMO: 727760

Lat: 47.4614N

Lon: 111.3847W

Elev: 3711

StdP: 12.83

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |             | Humidification DP/MCDB and HR |            |              |              |            |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|--------------|-------------|-------------------------------|------------|--------------|--------------|------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|------------|--------------|
|               |              |             | 99.6%                         |            |              | 99%          |            |             | 0.4%                  |             | 1%          |             |                       |            |              |
|               | 99.6%        | 99%         | DP                            | HR         | MCDB         | DP           | HR         | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-15.1 | (c)<br>-8.9 | (d)<br>-20.7                  | (e)<br>2.0 | (f)<br>-14.6 | (g)<br>-14.2 | (h)<br>2.9 | (i)<br>-8.5 | (j)<br>36.0           | (k)<br>43.8 | (l)<br>33.3 | (m)<br>42.6 | (n)<br>6.0            | (o)<br>220 | (p)<br>0.702 |

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>30.7            | (c)<br>92.9     | (d)<br>60.9 | (e)<br>89.5 | (f)<br>60.1 | (g)<br>86.1 | (h)<br>59.3 | (i)<br>64.0         | (j)<br>84.6 | (k)<br>62.3 | (l)<br>82.6 | (m)<br>60.9 | (n)<br>80.8 | (o)<br>10.5          | (p)<br>220 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 57.7                            | 81.7 | 67.4 | 55.5 | 75.2 | 67.0 | 53.4 | 69.6 | 66.1 | 31.0          | 84.7 | 29.7 | 81.8 | 28.7 | 80.8 | 75.2              |

Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |      |                    | n-Year Return Period Values of Extreme Temperature |           |       |            |       |            |       |            |       |
|-------------------|------|------|----------------------------|-------|------|--------------------|----------------------------------------------------|-----------|-------|------------|-------|------------|-------|------------|-------|
| 1%                |      | 2.5% | 5%                         | Mean  |      | Standard Deviation |                                                    | n=5 years |       | n=10 years |       | n=20 years |       | n=50 years |       |
| Min               |      | Max  |                            | Min   | Max  | Min                | Max                                                | Min       | Max   | Min        | Max   | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c)  |                            | (e)   | (f)  | (g)                | (h)                                                | (i)       | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 31.1              | 27.0 | 24.5 | DB                         | -23.7 | 98.4 | 6.2                | 3.1                                                | -28.1     | 100.6 | -31.7      | 102.4 | -35.2      | 104.2 | -39.7      | 106.4 |
|                   |      |      | WB                         | -23.7 | 67.7 | 5.9                | 2.2                                                | -28.0     | 69.3  | -31.4      | 70.5  | -34.7      | 71.7  | -39.0      | 73.3  |

Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 45.2   | 25.9  | 26.1  | 34.3  | 42.7  | 51.7  | 59.8  | 69.1  | 67.3  | 57.9  | 45.4  | 34.3  | 25.9  |
|                                                                   | DBStd       | 19.05  | 15.58 | 15.22 | 13.35 | 9.36  | 8.14  | 6.96  | 6.40  | 6.61  | 9.00  | 10.13 | 13.69 | 14.80 |
|                                                                   | HDD50       | 3711   | 748   | 671   | 497   | 249   | 76    | 8     | 0     | 0     | 30    | 203   | 482   | 747   |
|                                                                   | HDD65       | 7593   | 1211  | 1089  | 953   | 668   | 416   | 184   | 33    | 53    | 246   | 608   | 920   | 1212  |
|                                                                   | CDD50       | 1942   | 1     | 1     | 9     | 31    | 129   | 301   | 593   | 538   | 266   | 61    | 12    | 0     |
|                                                                   | CDD65       | 350    | 0     | 0     | 0     | 0     | 4     | 27    | 161   | 126   | 31    | 1     | 0     | 0     |
|                                                                   | CDH74       | 5789   | 0     | 0     | 0     | 12    | 137   | 506   | 2511  | 2010  | 587   | 26    | 0     | 0     |
|                                                                   | CDH80       | 2426   | 0     | 0     | 0     | 1     | 21    | 153   | 1166  | 876   | 205   | 4     | 0     | 0     |
|                                                                   | WSAvg       | 11.2   | 13.4  | 11.9  | 11.9  | 11.1  | 10.2  | 9.7   | 8.9   | 8.9   | 9.8   | 11.8  | 13.4  | 13.5  |
|                                                                   | PrecAvg     | 14.3   | 0.7   | 0.5   | 0.9   | 1.4   | 2.3   | 2.5   | 1.3   | 1.5   | 1.2   | 0.8   | 0.6   | 0.6   |
| Precipitation                                                     | PrecMax     | 23.4   | 1.9   | 1.2   | 2.1   | 4.3   | 5.2   | 6.9   | 4.7   | 4.9   | 3.2   | 3.2   | 1.7   | 1.9   |
|                                                                   | PrecMin     | 9.1    | 0.0   | 0.0   | 0.1   | 0.0   | 0.5   | 0.5   | 0.0   | 0.0   | 0.1   | 0.0   | 0.1   | 0.0   |
|                                                                   | PrecStd     | 3.4    | 0.5   | 0.3   | 0.6   | 0.9   | 1.1   | 1.5   | 1.1   | 1.2   | 0.8   | 0.6   | 0.4   | 0.5   |
|                                                                   |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 56.4  | 59.2  | 67.2  | 75.7  | 83.1  | 90.2  | 98.2  | 96.2  | 90.9  | 77.5  | 66.8  | 55.1  |
|                                                                   |             | MCWB   | 41.6  | 43.3  | 47.0  | 52.0  | 57.6  | 62.0  | 63.4  | 61.1  | 59.1  | 53.5  | 48.2  | 41.9  |
|                                                                   | 2%          | DB     | 51.5  | 53.4  | 61.9  | 69.7  | 78.0  | 84.4  | 94.1  | 92.1  | 86.0  | 71.4  | 60.2  | 50.8  |
|                                                                   |             | MCWB   | 40.0  | 40.9  | 44.8  | 49.8  | 55.7  | 60.6  | 61.9  | 59.9  | 57.7  | 51.1  | 45.4  | 39.5  |
|                                                                   | 5%          | DB     | 47.8  | 49.3  | 57.1  | 64.1  | 73.3  | 79.7  | 90.6  | 88.8  | 81.2  | 66.0  | 55.6  | 47.1  |
|                                                                   |             | MCWB   | 37.8  | 38.3  | 42.7  | 47.4  | 53.8  | 58.8  | 60.7  | 59.3  | 56.2  | 49.2  | 42.6  | 37.5  |
|                                                                   | 10%         | DB     | 44.4  | 45.2  | 52.5  | 59.0  | 68.2  | 75.5  | 86.9  | 85.2  | 75.9  | 60.9  | 51.7  | 43.5  |
|                                                                   |             | MCWB   | 35.9  | 36.1  | 40.3  | 44.6  | 51.7  | 57.1  | 59.9  | 58.8  | 54.6  | 46.8  | 40.8  | 35.0  |
|                                                                   | 0.4%        | WB     | 43.7  | 44.8  | 48.2  | 53.9  | 59.4  | 65.7  | 66.8  | 64.3  | 62.2  | 55.9  | 49.7  | 43.3  |
|                                                                   |             | MCDB   | 53.3  | 56.7  | 64.3  | 71.7  | 77.9  | 82.5  | 87.1  | 86.4  | 81.2  | 74.1  | 63.7  | 53.5  |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 40.5  | 41.4  | 45.8  | 50.7  | 56.9  | 62.5  | 64.8  | 62.4  | 59.5  | 52.4  | 46.1  | 40.6  |
|                                                                   |             | MCDB   | 50.2  | 52.0  | 60.4  | 66.9  | 74.7  | 79.3  | 86.5  | 84.3  | 79.9  | 68.1  | 58.8  | 49.6  |
|                                                                   | 5%          | WB     | 38.3  | 38.8  | 43.1  | 48.1  | 54.9  | 60.4  | 63.1  | 61.1  | 57.5  | 50.0  | 43.5  | 37.9  |
|                                                                   |             | MCDB   | 46.9  | 48.4  | 56.2  | 62.9  | 70.3  | 76.3  | 84.0  | 82.8  | 78.1  | 64.1  | 54.4  | 46.1  |
|                                                                   | 10%         | WB     | 36.1  | 36.3  | 40.7  | 45.5  | 52.8  | 58.5  | 61.6  | 59.8  | 55.3  | 47.6  | 41.2  | 35.5  |
|                                                                   |             | MCDB   | 43.7  | 44.9  | 51.7  | 57.9  | 66.3  | 72.9  | 81.9  | 80.8  | 74.0  | 60.2  | 51.0  | 42.9  |
| Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 18.0  | 19.0  | 21.4  | 22.4  | 23.5  | 24.7  | 30.7  | 30.3  | 26.4  | 21.8  | 18.0  | 17.0  |
|                                                                   |             | MCDBR  | 19.2  | 22.5  | 27.0  | 31.0  | 31.6  | 31.8  | 35.9  | 36.3  | 34.9  | 29.0  | 21.8  | 18.4  |
|                                                                   |             | MCWBR  | 11.9  | 13.4  | 14.6  | 15.7  | 14.2  | 13.2  | 12.7  | 13.1  | 14.5  | 14.2  | 11.8  | 11.6  |
|                                                                   | 5% WB       | MCDBR  | 18.4  | 21.7  | 25.8  | 29.3  | 28.5  | 28.8  | 32.0  | 32.4  | 32.5  | 26.9  | 21.7  | 18.3  |
|                                                                   |             | MCWBR  | 12.1  | 13.4  | 14.4  | 15.3  | 13.3  | 13.3  | 12.6  | 12.2  | 14.1  | 13.8  | 12.7  | 12.3  |
|                                                                   |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
| Clear-Sky Solar Irradiance                                        | taub        |        | 0.247 | 0.256 | 0.272 | 0.315 | 0.326 | 0.327 | 0.331 | 0.348 | 0.312 | 0.282 | 0.255 | 0.242 |
|                                                                   | taud        |        | 2.508 | 2.519 | 2.545 | 2.422 | 2.441 | 2.477 | 2.480 | 2.391 | 2.469 | 2.554 | 2.539 | 2.497 |
|                                                                   | Ebn at Noon |        | 277   | 297   | 306   | 298   | 298   | 296   | 294   | 284   | 286   | 282   | 271   | 264   |
|                                                                   | Edh at Noon |        | 19    | 23    | 27    | 34    | 35    | 34    | 33    | 35    | 29    | 23    | 18    | 17    |
| All-Sky Solar Radiation                                           | RadAvg      |        | 462   | 775   | 1153  | 1534  | 1813  | 2093  | 2288  | 1892  | 1364  | 863   | 512   | 368   |
|                                                                   | RadStd      |        | 30    | 52    | 90    | 87    | 163   | 141   | 116   | 105   | 104   | 75    | 42    | 26    |

Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (12 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page